

ELEC5305 Project Proposal

Detection and Classification of Power Quality Disturbances Using Time-Frequency Signal Analysis and Machine Learning

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1. Project Overview

Power quality disturbances can affect the reliable operation of electrical equipment, industrial systems, and sensitive electronic loads. Common power quality disturbances include voltage sags, voltage swells, voltage interruptions, harmonics, and transient events. The goal of this project is to develop a signal-processing-based system capable of automatically detecting and classifying various common power quality disturbances from voltage waveforms.

The proposed system will integrate time-domain analysis, time-frequency domain signal processing, and machine learning classification methods. The project will investigate three signal analysis techniques: Fast Fourier Transform (FFT), Short-Time Fourier Transform (STFT), and Discrete Wavelet Transform (DWT). The system will extract features from these diverse signal representations and utilize them to train classifiers, such as Support Vector Machines (SVM) and Random Forests. Ultimately, the performance of different feature representation methods and classifiers will be compared based on metrics like classification accuracy and robustness to noise, to determine the more suitable approach for automatic identification of power quality disturbances.

2. Background and Motivation

The modern power system contains an increasing number of nonlinear loads, power electronic converters, renewable energy interfaces, and sensitive digital devices. These components can cause power quality issues and may also be affected by them. Therefore, automatic identification of abnormal voltage events is of great significance for monitoring and diagnosis. Traditional Fourier analysis is effective in identifying steady-state frequency components (such as harmonics), but provides limited information on the timing of short-term disturbances. Short time Fourier transform (STFT) achieves time-domain localization by applying Fourier transform on a short-time window. Wavelet analysis provides multi-resolution representation, particularly suitable for detecting mutations and non-stationary events. Previous studies have shown that wavelet features can be used for power quality assessment, and time-frequency features can be combined with machine learning classifiers such as support vector machines (SVM) [2] - [5].

3. Proposed Methodology

The project will be implemented primarily in Python using NumPy, SciPy, PyWavelets, scikit-learn, Matplotlib, and Jupiter Notebook. MATLAB may also be used to verify signal generation and selected DSP results.

First, a dataset of voltage waveforms will be created using mathematical models of a 50 Hz power signal. Six classes will be considered:

- Normal signal
- Voltage sag

- Voltage swell
- Voltage interruption
- Harmonic distortion
- Transient disturbance

For each class, parameters such as disturbance magnitude, start time, duration, phase, and noise level will be randomly varied so that the classifier is not trained on identical waveforms. The disturbance categories will be designed with reference to established power-quality monitoring definitions [1]. If a suitable public dataset is available, it may also be used as an additional validation dataset.

Second, each waveform will be pre-processed and analyzed using three signal-processing approaches.

FFT analysis will be used to examine the global frequency spectrum and identify harmonic components.

STFT analysis will generate time-frequency spectrograms showing how the frequency content of a signal changes during a disturbance.

DWT analysis will decompose each signal into multiscale approximation and detail coefficients, allowing short-duration and non-stationary events to be represented more effectively.

Candidate features will include RMS voltage, peak value, total harmonic distortion, spectral energy, dominant-frequency components, STFT band energy, and wavelet sub-band energy.

Third, the extracted features will be normalized and divided into training and testing datasets. An SVM will be used as the main classifier, while Random Forest will provide a second classification method for comparison. Experiments will compare FFT-based features, STFT-based features, DWT-based features, and combined time-frequency features. Additional noise will also be introduced to selected test signals to evaluate robustness under different signal-to-noise conditions.

4. Expected Outcomes

The expected outcome is a working prototype that accepts a voltage waveform and predicts its corresponding power quality category. The system will provide time-domain voltage waveforms, FFT frequency spectra, STFT spectrograms, DWT coefficient distributions, and classification confusion matrices.

Classification performance will be evaluated using overall accuracy, precision, recall, F1-score, and confusion matrices. The effect of measurement noise will also be investigated at several signal-to-noise ratio levels. The final analysis will determine which signal representation provides the best balance between classification performance, computational complexity, noise robustness, and interpretability.

All source code, implementation instructions, representative experimental results, the project proposal, and the final demonstration will be documented in the public GitHub repository.

5. Timeline (Weeks 1-13)

1-2	Finalize project scope, disturbance classes, methodology, and GitHub repository structure.
3-5	Conduct literature review, define mathematical disturbance models, generate and inspect the dataset.
6-7	Implement FFT, STFT, and DWT analysis; develop visualizations and feature extraction.
8-9	Train SVM and Random Forest classifiers and perform initial experiments.
10-11	Optimize parameters, compare feature sets, evaluate noise robustness, and generate final results.
12-13	Complete final report, clean project code, prepare README, GitHub Pages, and final demonstration.

6. References

- [1] IEEE, IEEE Recommended Practice for Monitoring Electric Power Quality, IEEE Std 1159-2019, 2019.
- [2] S. Santoso, E. J. Powers, W. M. Grady, and P. Hofmann, "Power quality assessment via wavelet transform analysis," IEEE Transactions on Power Delivery, vol. 11, no. 2, pp. 924-930, 1996.
- [3] J. Wang, Z. Xu, and Y. Che, "Power Quality Disturbance Classification Based on DWT and Multilayer Perceptron Extreme Learning Machine," Applied Sciences, vol. 9, no. 11, Art. 2315, 2019.
- [4] Q. Tang, W. Qiu, and Y. Zhou, "Classification of Complex Power Quality Disturbances Using Optimized S-Transform and Kernel SVM," IEEE Transactions on Industrial Electronics, vol. 67, no. 11, pp. 9715-9723, 2020.
- [5] W.-M. Lin, C.-H. Wu, C.-H. Lin, and F.-S. Cheng, "Detection and Classification of Multiple Power-Quality Disturbances With Wavelet Multiclass SVM," IEEE Transactions on Power Delivery, vol. 23, no. 4, pp. 2575-2582, 2008.